Deploy

📂 Data Upload

Upload your business dataset.

Supported formats:

- 📦 ZIP
- 📄 CSV
- 📊 Excel (.xlsx)

Choose Dataset

Walmart.csv
355.2KB

✅ CSV Uploaded Successfully

Shape: `(6435, 8)`

Columns:

```
[
  0 : "Store"
  1 : "Date"
  2 : "Weekly_Sales"
  3 : "Holiday_Flag"
  4 : "Temperature"
  5 : "Fuel_Price"
  6 : "CPI"
  7 : "Unemployment"
]
```

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemploymen
0	1	05-02-2010	1643690.9	0	42.31	2.572	211.0964	8.10
1	1	12-02-2010	1641957.44	1	38.51	2.548	211.2422	8.10
2	1	19-02-2010	1611968.17	0	39.93	2.514	211.2891	8.10
3	1	26-02-2010	1409727.59	0	46.63	2.561	211.3196	8.10
4	1	05-03-2010	1554806.68	0	46.5	2.625	211.3501	8.10

Deploy

Data Profile

Dataset Understanding


```
{
  "type" : "sales"
  "date_columns" : []
  "numeric_columns" : [
    0 : "Store"
    1 : "Weekly_Sales"
    2 : "Holiday_Flag"
    3 : "Temperature"
    4 : "Fuel_Price"
    5 : "CPI"
    6 : "Unemployment"
  ]
  "categorical_columns" : [
    0 : "Date"
  ]
}
```


Semantic Model


```
{
  "date" : "Date"
  "revenue" : "Weekly_Sales"
  "price" : "Fuel_Price"
}
```


Generated Metrics


```
{
  "Total Records" : 6435
}
```


Dataset Intelligence


```
{
  "rows" : 6435
  "columns" : 8
  "numeric_columns" : [
    0 : "Store"
    1 : "Weekly_Sales"
    2 : "Holiday_Flag"
    3 : "Temperature"
    4 : "Fuel_Price"
    5 : "CPI"
    6 : "Unemployment"
  ]
  "categorical_columns" : [
    0 : "Date"
  ]
  "date_columns" : [
    0 : "Date"
  ]
  "binary_columns" : [
    0 : "Holiday_Flag"
  ]
  "id_columns" : [
    0 : "Holiday_Flag"
  ]
  "measure_columns" : [
    0 : "Store"
    1 : "Temperature"
    2 : "Fuel_Price"
    3 : "CPI"
    4 : "Unemployment"
  ]
  "dimension_columns" : []
}
```


Dataset loaded successfully

Preview

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemploymer
0	1	05-02-2010	1643690.9	0	42.31	2.572	211.0964	8.10
1	1	12-02-2010	1641957.44	1	38.51	2.548	211.2422	8.10
2	1	19-02-2010	1611968.17	0	39.93	2.514	211.2891	8.10
3	1	26-02-2010	1409727.59	0	46.63	2.561	211.3196	8.10
4	1	05-03-2010	1554806.68	0	46.5	2.625	211.3501	8.10

Dataset Profile

Rows	Columns	Missing Values	Duplicates
6435	8	0	0

Memory (MB)	Numeric Columns	Categorical Columns	Date Columns
0.45	7	1	1

Column Information

	Column	Data Type	Missing Values	Unique Values
Store	Store	int64	0	45
Date	Date	str	0	143
Weekly_Sales	Weekly_Sales	float64	0	6435
Holiday_Flag	Holiday_Flag	int64	0	2
Temperature	Temperature	float64	0	3528
Fuel_Price	Fuel_Price	float64	0	892
CPI	CPI	float64	0	2145
Unemployment	Unemployment	float64	0	349

Dataset Summary

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
count	6435	6435	6435	6435	6435	6435	6435	6435
unique	None	143	None	None	None	None	None	None
top	None	05-02-2	None	None	None	None	None	None
freq	None	45	None	None	None	None	None	None
mean	23	None	1046964.8776	0.0699	60.6638	3.3586	171.5784	7.9992
std	12.9882	None	564366.6221	0.255	18.4449	0.459	39.3567	1.8759
min	1	None	209986.25	0	-2.06	2.472	126.064	3.879
25%	12	None	553350.105	0	47.46	2.933	131.735	6.891
50%	23	None	960746.04	0	62.67	3.445	182.6165	7.874
75%	34	None	1420158.66	0	74.94	3.735	212.7433	8.622

Missing Values

No missing values found.

Deploy

- app
- Data Upload
- Data Profile
- **Business Dashboard**
- AI Insights
- Ask Your Data
- Forecasting
- Executive Report

📊 Business Dashboard

Detected Dataset Type: GENERIC

🔎 Dashboard Data Check

Shape: `(6435, 8)`

Columns:


```
[
  0 : "Store"
  1 : "Date"
  2 : "Weekly_Sales"
  3 : "Holiday_Flag"
  4 : "Temperature"
  5 : "Fuel_Price"
  6 : "CPI"
  7 : "Unemployment"
]
```

📈 Business KPIs

📄 Total Records	💰 Total Value	📊 Average Value	💲 Average Price
6435	6737218987.11	1046964.88	3.36

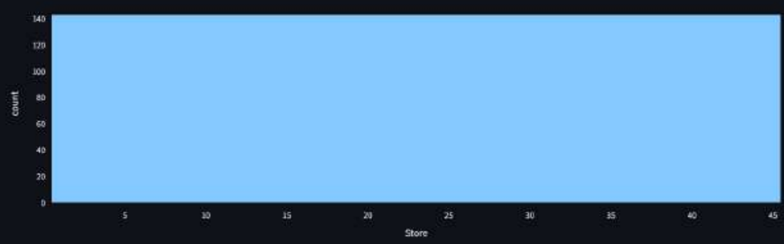
❌ Missing Values

0

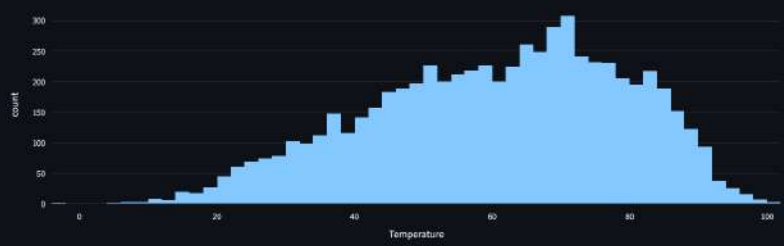
📊 Business Visualizations

Store Trend

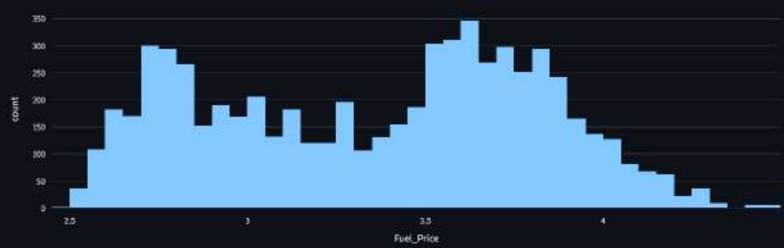
Store Distribution



Temperature Distribution



Fuel_Price Distribution



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Ask Your Data

Ask a question about your data

explain the dataset

Certainly, I can help you understand this dataset. As a Business Intelligence Analyst, I see this as a valuable dataset for understanding retail sales performance and its various influencing factors.

Let's break down each column and discuss its potential business implications:

Dataset Explanation & Business Insights:

1. **Store:**
   - **Explanation:** This column represents a unique identifier for each individual retail store.
   - **Business Insight:** This is crucial for **store-specific performance analysis**. You can compare sales trends, profitability, and the impact of external factors across different stores. It allows for identifying top-performing stores, underperforming stores, and understanding regional or demographic differences in consumer behavior. For instance, you could calculate the average weekly sales for each store to benchmark their performance.
2. **Date:**
   - **Explanation:** This column indicates the specific week for which the sales and other data points are recorded.
   - **Business Insight:** The 'Date' column is fundamental for **time-series analysis**. You can observe sales trends over time, identify seasonality (e.g., higher sales during specific months or holidays), analyze year-over-year growth, and detect any long-term patterns or shifts in consumer behavior. This is vital for forecasting future sales and planning inventory.
3. **Weekly_Sales:**
   - **Explanation:** This is your primary target variable, representing the total sales revenue for a given store in that particular week.
   - **Business Insight:** This is a core **Key Performance Indicator (KPI)** for any retail business. All other variables can be analyzed in relation to Weekly_Sales to understand their impact. You can track sales growth, identify periods of high or low sales, analyze sales volatility, and ultimately measure the success of marketing campaigns or operational changes.
4. **Holiday_Flag:**
   - **Explanation:** This is a binary flag (0 or 1) indicating whether the week includes a special holiday (e.g., Super Bowl, Thanksgiving, Christmas, Easter). A value of '1' signifies a holiday week, '0' signifies a regular week.
   - **Business Insight:** This is incredibly useful for understanding the **impact of holidays on sales**. You can compare sales performance during holiday weeks versus non-holiday weeks to quantify the boost or dip in demand. This information is critical for promotional planning, inventory stocking, staffing adjustments, and marketing strategies around peak holiday seasons.
5. **Temperature:**
   - **Explanation:** This column records the average temperature for the week in the region of the store.
   - **Business Insight:** Temperature can influence consumer behavior and purchasing decisions, especially for certain product categories. For example, higher temperatures might lead to increased sales of beverages, ice cream, or summer clothing, while lower temperatures could boost sales of comfort food, hot drinks, or winter apparel. Analyzing its correlation with sales can help in **optimizing product assortments and promotions** based on weather forecasts.
6. **Fuel_Price:**
   - **Explanation:** This represents the average fuel price in the region for that week.
   - **Business Insight:** Fuel prices are an important economic indicator that can impact both consumer spending and operational costs. Higher fuel prices can reduce consumers' disposable income, potentially leading to a decrease in non-essential purchases. It can also increase logistical costs for the business. Monitoring this can provide insights into **consumer purchasing power and potential cost pressures**.
7. **CPI (Consumer Price Index):**
   - **Explanation:** This measures the average change over time in the prices paid by urban consumers for a market basket of consumer goods and services. It's a key indicator of inflation.
   - **Business Insight:** CPI provides a macro-economic view of **inflation and purchasing power**. A rising CPI generally means goods and services are more expensive, which can affect consumer confidence and their ability to spend on retail products. Analyzing its trend alongside sales can help understand the broader economic environment's influence on your business.
8. **Unemployment:**
   - **Explanation:** This represents the unemployment rate in the region for that week.
   - **Business Insight:** The unemployment rate is another critical **economic health indicator**. Higher unemployment typically leads to reduced consumer spending due to lower overall income and decreased consumer confidence. Conversely, lower unemployment can stimulate spending. Understanding its relationship with sales helps gauge the economic well-being of your customer base and adjust strategies accordingly.

Overall Business Value:

This dataset offers a comprehensive view of factors influencing retail sales. By analyzing these variables together, you can:

- **Identify key sales drivers:** Determine which factors (holidays, temperature, economic conditions) have the most significant impact on sales.
- **Improve sales forecasting:** Develop more accurate models by incorporating these external variables.
- **Optimize marketing and promotions:** Target specific weeks or conditions with relevant campaigns.
- **Enhance inventory management:** Stock products strategically based on anticipated demand linked to holidays, weather, or economic forecasts.
- **Understand economic resilience:** Assess how your stores perform under varying economic conditions.

In essence, this dataset enables a deeper, data-driven understanding of your retail operations, empowering more informed strategic and tactical business decisions.

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🧠 AI Insights

Gemini Business Insights

Here's a business analysis based on the provided dataset:

Dataset Overview:

- **Total Records:** 6435
- **Missing Values:** 0
- The dataset contains weekly sales data for multiple stores, along with contextual information such as date, holiday status, temperature, fuel price, consumer price index (CPI), and unemployment rate. The absence of missing values indicates a high level of data completeness for the recorded attributes.

---

1. Key Business Insights

- **Fact:** The dataset provides granular weekly sales (`Weekly_Sales`) data across different stores (`Store`).
  - **Possible interpretation:** This allows for detailed performance tracking at both the store and aggregated levels.
- **Fact:** Each sales record is associated with a specific `Date` , `Holiday_Flag` , `Temperature` , `Fuel_Price` , `CPI` , and `Unemployment` rate.
  - **Possible interpretation:** This comprehensive set of attributes enables analysis of how various internal and external factors might influence sales performance.
- **Fact:** There are 0 missing values across all columns.
  - **Possible interpretation:** This indicates excellent data quality and integrity for the collected information, minimizing the need for data cleaning or imputation prior to analysis.
- **Fact:** The `Holiday_Flag` explicitly marks weeks that are holidays.
  - **Possible interpretation:** This allows for direct comparison of sales performance during holiday versus non-holiday periods, which is crucial for understanding demand patterns.

---

2. Important Trends

- **Fact:** The presence of the `Date` column allows for time-series analysis.
  - **Possible interpretation:** Trends in `Weekly_Sales` (e.g., seasonality, year-over-year growth, or decline) can be identified over the period covered by the dataset.
- **Fact:** Trends in external factors like `Temperature` , `Fuel_Price` , `CPI` , and `Unemployment` can be observed over time.
  - **Possible interpretation:** Analyzing these trends alongside `Weekly_Sales` can reveal correlations and potential causal relationships between economic, environmental, and sales performance. For instance, seasonal temperature shifts might correlate with specific product sales cycles, or rising fuel prices might impact consumer spending or operational costs.
- **Fact:** `Holiday_Flag` allows for specific trend analysis related to holiday periods.
  - **Possible interpretation:** It can be observed whether sales consistently increase or decrease during holiday weeks compared to non-holiday weeks, and if this trend varies by store or over different years.
- **Limitation:** Specific trend directions (e.g., "sales are growing by X%") or magnitudes of change are not available from the provided KPIs and require further calculation based on the raw data.

---

3. Risks or Problems

- **Fact:** `Weekly_Sales` are influenced by external factors such as `Temperature` , `Fuel_Price` , `CPI` , and `Unemployment` .
  - **Possible interpretation:** Adverse fluctuations in these external factors, such as a significant increase in `Unemployment` or `Fuel_Price` , or extreme `Temperature` conditions, could pose a risk to sales stability and overall revenue.
- **Fact:** The dataset contains data from multiple `Store` locations over time.
  - **Possible interpretation:** Without specific analysis, potential inconsistencies in sales performance across different stores or periods of underperformance in certain locations could be a hidden risk. Identifying these would require comparative analysis between stores.
- **Fact:** `Holiday_Flag` marks specific high-impact weeks.
  - **Possible interpretation:** Inaccurate forecasting or inadequate preparation (e.g., inventory management, staffing) for holiday weeks could lead to missed sales opportunities or increased operational inefficiencies. However, the extent of this risk is not observable from the provided KPIs.
- **Limitation:** Specific problems such as declining sales, high volatility, or underperforming stores cannot be identified without performing calculations on the dataset beyond the provided KPIs.

---

4. Business Recommendations

Based on the available dataset structure and insights:

1. **Conduct Comprehensive Sales Performance Analysis:**
   - Analyze `Weekly_Sales` over time to identify seasonality, long-term trends, and high/low sales periods.
   - Break down `Weekly_Sales` by `Store` to identify top-performing, average, and underperforming locations, allowing for targeted operational improvements or best practice sharing.
2. **Quantify the Impact of External Factors:**
   - Perform correlation and regression analysis to understand the quantitative relationship between `Weekly_Sales` and `Temperature` , `Fuel_Price` , `CPI` , and `Unemployment` . This can help in building more accurate sales forecasting models.
   - Specifically, analyze how significant changes in `Fuel_Price` or `Unemployment` rates impact consumer spending patterns reflected in `Weekly_Sales` .
3. **Optimize Holiday Sales Strategies:**
   - Analyze the average `Weekly_Sales` during `Holiday_Flag = 1` weeks versus `Holiday_Flag = 0` weeks to quantify the sales uplift (or potential dip) attributable to holidays.
   - Use this insight to optimize inventory levels, staffing, and promotional activities specifically for holiday periods to maximize revenue and customer satisfaction.
4. **Proactive Economic Monitoring:**
   - Establish a process to continuously monitor trends in `CPI` and `Unemployment` as early indicators of potential shifts in consumer purchasing power. This can help in proactively adjusting sales strategies or promotional campaigns.
5. **Utilize Data Quality:**
   - Leverage the 0 missing values to ensure that all analyses are based on complete information, reducing the complexity and potential errors associated with data imputation.

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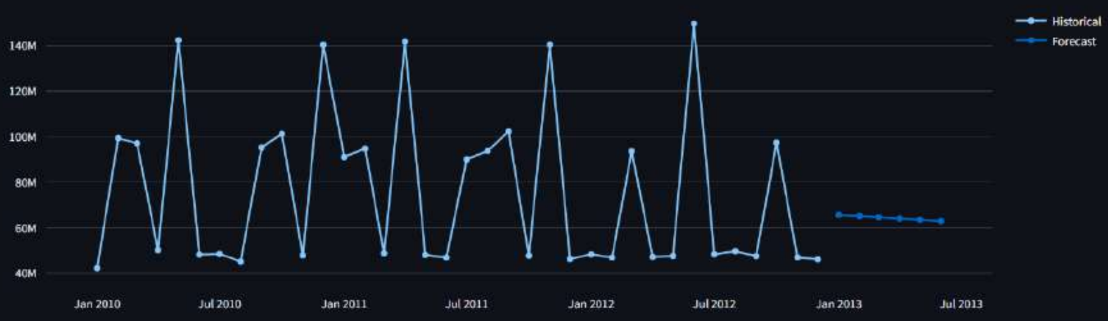
📈 Business Forecasting

📊 Forecast Capability

Forecast Available

Detected Dataset: GENERIC

📈 Weekly_Sales Forecast



📌 Forecast Summary

Current Value
46128514.25

Future Prediction
62968872.71

Expected Change
36.51%

Forecast trend: Increasing

🤖 AI Forecast Explanation

Here is a short business forecast explanation:

1. Forecast Observation The historical last recorded value was 46,128,514.25. The future predicted value is 62,968,872.71, representing an expected increase of 36.51%.

2. Trend Explanation The forecast indicates a clear increasing trend, driven by the substantial growth projected from the historical last value to the future predicted value.

3. Assumptions and Limitations This forecast is based solely on the provided numerical data and observed trend. It assumes that the underlying factors driving past performance will continue to contribute positively. External factors, market conditions, and specific operational drivers that might influence this change have not been provided and are therefore not considered in this explanation.

🔮 Future Prediction

	TimeIndex	ForecastValue	Date
0	36	65638448.0062	2013-01-01 00:00:00
1	37	65104532.9464	2013-02-01 00:00:00
2	38	64570617.8867	2013-03-01 00:00:00
3	39	64036702.827	2013-04-01 00:00:00
4	40	63502787.7672	2013-05-01 00:00:00
5	41	62968872.7075	2013-06-01 00:00:00

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AI Insights
Ask Your Data
Forecasting
Executive Report

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📄 Executive Business Report

Detected Dataset Type: GENERIC

📊 Business Overview

📄 Total Records	💰 Total Value	📊 Average Value	💲 Average Price
6435	6737218987.11	1046964.88	3.36

❌ Missing Values
0

🤖 Executive Summary

Executive Report: Business Operations Overview

1. Overall Business Performance

This report provides a high-level overview of business operations based on the provided generic dataset and aggregated Key Performance Indicators (KPIs). The dataset comprises 6,435 weekly records across various stores.

- **Total Operational Value:** The aggregate operational value recorded across all stores and weeks is **$6,737,218,987.11**. This figure represents the sum of weekly operational activities over the entire dataset period.
- **Average Weekly Operational Value:** On average, each store recorded a weekly operational value of **$1,046,964.88**. This provides a benchmark for typical weekly activity levels.
- **Average Price:** An "Average Price" KPI is provided at **$3.36**. Without further context from the dataset, the specific item or service this price refers to is not clear, but it likely represents an average unit price associated with the recorded weekly operational value.
- **Data Quality:** The dataset exhibits excellent data quality with **0 missing values** across all recorded attributes, ensuring the reliability of the aggregated metrics.

2. Key Trends and Patterns

Based solely on the provided aggregate KPIs, detailed time-series trends, patterns, and correlations cannot be precisely calculated or described. However, the presence of specific columns in the dataset suggests potential areas for future in-depth analysis:

- **Time-Based Analysis:** The 'Date' column indicates that time-series analysis is possible. Such analysis, with access to the full dataset, could reveal weekly, monthly, or yearly trends in 'Weekly Operational Value', identify seasonality, growth rates, or the impact of specific periods (e.g., holidays).
- **External Factor Influence:** The dataset includes 'Temperature', 'Fuel_Price', 'CPI' (Consumer Price Index), and 'Unemployment'. These external factors are typically influential in various business contexts. A comprehensive analysis of the full dataset would be required to determine if and how these factors correlate with or drive fluctuations in 'Weekly Operational Value'.
- **Holiday Impact:** The 'Holiday_Flag' column suggests that operational value might be influenced by holiday periods. Deeper analysis is needed to quantify this impact.
- **Store-Level Performance:** The 'Store' column indicates that performance can be analyzed at an individual store level, allowing for identification of top-performing or underperforming locations. This level of detail is not available from the aggregate KPIs.

3. Major Risks or Concerns

While the dataset shows good data integrity, several potential risks and concerns are suggested by the available columns:

- **Sensitivity to External Economic Factors:** The inclusion of 'CPI' and 'Unemployment' indicates a potential sensitivity of operational performance to broader economic conditions. Negative shifts in these indicators could pose a risk to future operational values. The extent of this sensitivity is Not available in the dataset without further analysis.
- **Dependency on Environmental/Supply Chain Variables:** The 'Temperature' and 'Fuel_Price' columns suggest that operational activities might be influenced by environmental factors or supply chain costs. Fluctuations in these areas could impact profitability or operational efficiency, even if the 'Weekly Operational Value' remains stable.
- **Lack of Granular Insights:** The current high-level KPIs do not allow for detailed diagnostics. Without understanding the specific drivers behind the 'Weekly Operational Value' (e.g., volume vs. price, specific operational segments), it is challenging to identify root causes for performance fluctuations or emerging issues.
- **Operational Volatility (Possible interpretation):** Weekly operational values can inherently be volatile. While not quantifiable with the provided KPIs, significant week-to-week variations could pose planning and resource allocation challenges.

4. Strategic Recommendations

To gain deeper insights and inform strategic decision-making, the following recommendations are proposed:

- **Conduct Comprehensive Data Analysis:** Initiate a detailed analytical project using the full dataset to uncover specific trends, correlations, and performance drivers. This analysis should focus on:
  - **Time-Series Analysis:** To understand seasonal patterns, growth trajectories, and the impact of holidays on 'Weekly Operational Value'.
  - **Factor Correlation Analysis:** To quantify the relationship between 'Weekly Operational Value' and external factors (CPI, Unemployment, Temperature, Fuel_Price). This will help in forecasting and risk mitigation.
  - **Store-Level Performance Benchmarking:** To identify high-performing stores and best practices, as well as areas requiring improvement.
- **Establish Key Performance Indicators (KPIs) Monitoring:** Implement a dashboard or regular reporting system to continuously monitor 'Weekly Operational Value' alongside key external factors. This proactive monitoring will enable timely responses to emerging trends or risks.
- **Explore Granular Data:** Investigate the underlying components of 'Weekly Operational Value'. If possible, break down this value into its constituent parts (e.g., number of transactions, average value per transaction) to provide more actionable insights.
- **Strategic Planning Integration:** Utilize the insights derived from deeper analysis to inform strategic planning cycles, including resource allocation, operational adjustments, and proactive risk management strategies related to economic and environmental fluctuations.

SEMANTIC CHECK


```
{
  "date" : "Date"
  "price" : "Fuel_Price"
  "revenue" : "Weekly_Sales"
  "external_factors" : "Temperature"
  "forecast_target" : "Weekly_Sales"
}
```


📈 Forecast: Weekly_Sales

🔮 Future Prediction

	Date	ForecastValue
0	2013-01-01 00:00:00	65638448.0062
1	2013-02-01 00:00:00	65104532.9464
2	2013-03-01 00:00:00	64570617.8867
3	2013-04-01 00:00:00	64036702.827
4	2013-05-01 00:00:00	63502787.7672
5	2013-06-01 00:00:00	62968872.7075